



计算机体系结构

3.3 TPU 和 NPU 以及大规模集群

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3.3 TPU 和 NPU 以及大规模集群

- 1、Google TPU
- 2、Huawei NPU
- 3、大规模集群

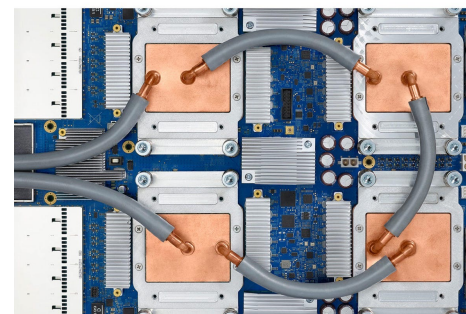
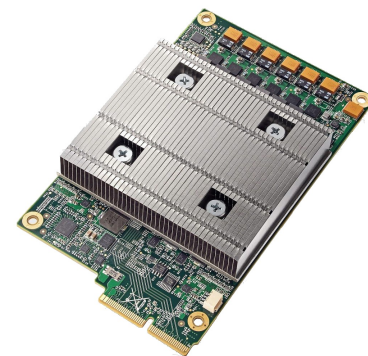


1、Google Tensor Processing Unit (TPU)

- TPU是 Google 开发的专用集成电路（ASIC），专门用于加速机器学习
- 只能在Google Cloud上用，并需要TensorFlow/JAX/PyTorch等框架

	TPUv1	TPUv2	TPUv3	TPUv4 ^{[16][18]}	TPUv5e ^[19]	TPUv5p ^{[20][21]}	v6e (Trillium) ^{[22][23]}
Date introduced	2015	2017	2018	2021	2023	2023	2024
Process node	28 nm	16 nm	16 nm	7 nm	Unstated	Unstated	
Die size (mm ²)	331	< 625	< 700	< 400	300-350	Unstated	
On-chip memory (MiB)	28	32	32	32	48	112	
Clock speed (MHz)	700	700	940	1050	Unstated	1750	
Memory	8 GiB DDR3	16 GiB HBM	32 GiB HBM	32 GiB HBM	16 GB HBM	95 GB HBM	32 GB
Memory bandwidth	34 GB/s	600 GB/s	900 GB/s	1200 GB/s	819 GB/s	2765 GB/s	1640 GB/s
TDP (W)	75	280	220	170	Not Listed	Not Listed	
TOPS (Tera Operations Per Second)	23	45	123	275	197 (bf16) 393 (int8)	459 (bf16) 918 (int8)	918 (bf16) 1836 (int8)
TOPS/W	0.31	0.16	0.56	1.62	Not Listed	Not Listed	

Source: https://en.wikipedia.org/wiki/Tensor_Processing_Unit

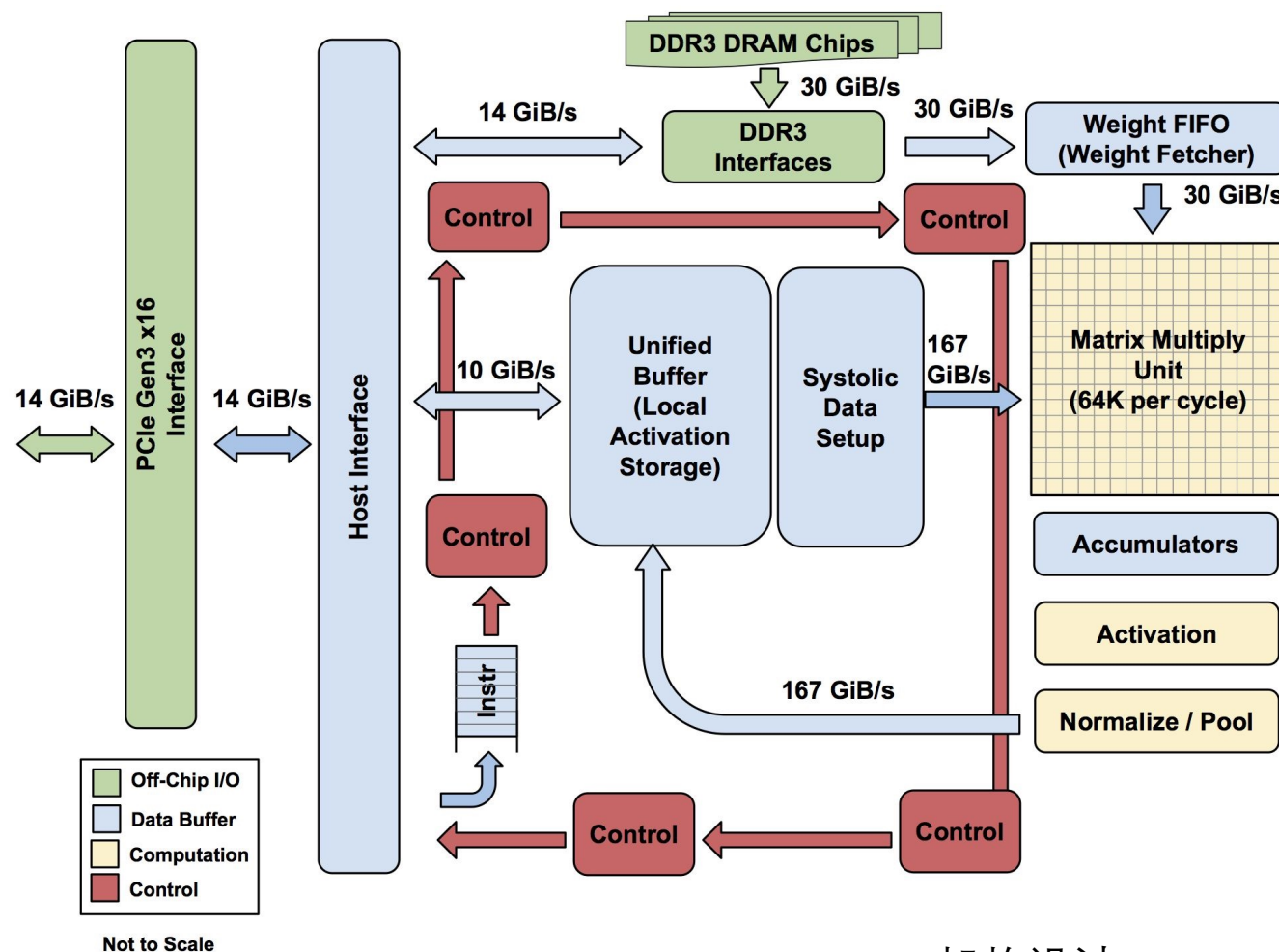


TPUv1架构



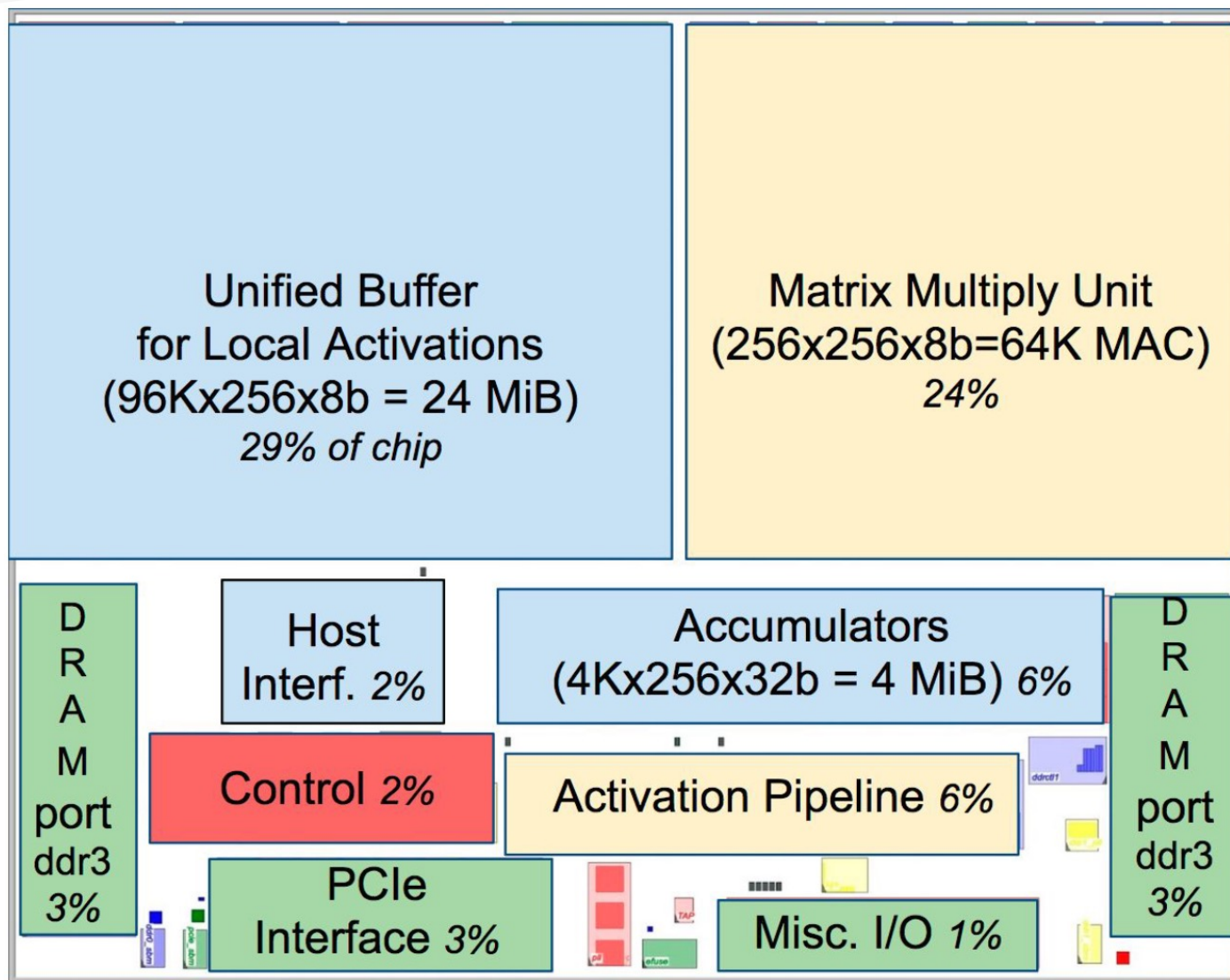
➤ 为推理设计，低Latency为目标

- ❑ The Matrix Unit: 65,536 (256x256) 8-bit multiply-accumulate units
- ❑ 700 MHz clock rate
- ❑ Peak: 92T operations/second :
 $65,536 * 2 * 700M$
- ❑ **24 MiB of on-chip Unified Buffer (activation memory)**
- ❑ Two 2133MHz DDR3 DRAM channels
- ❑ **8 GiB of off-chip weight DRAM memory**

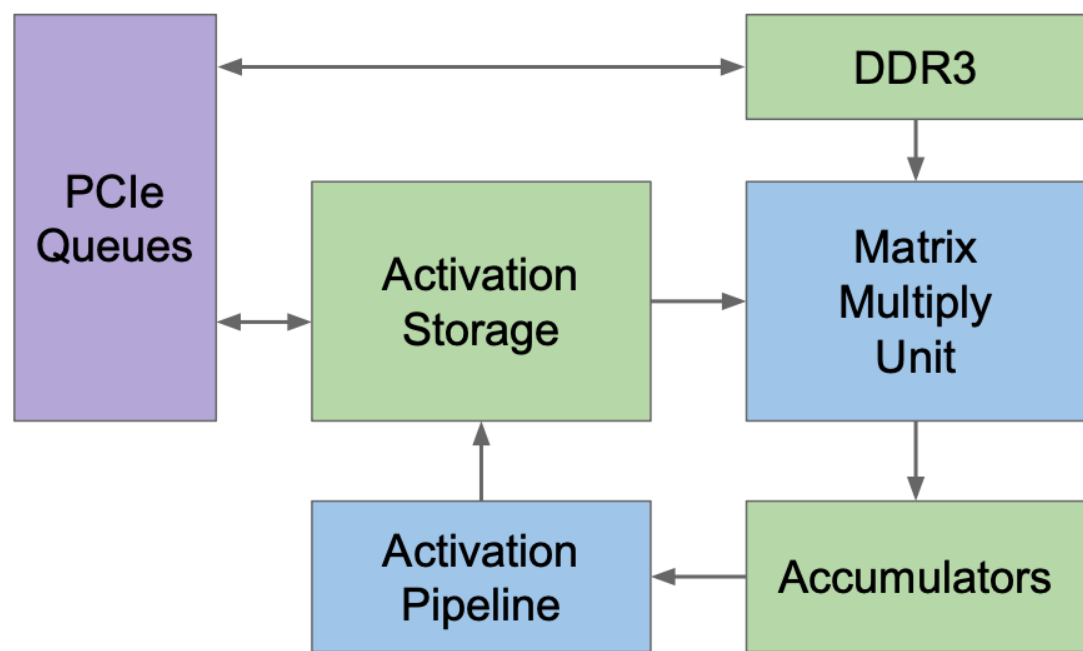


TPU架构设计

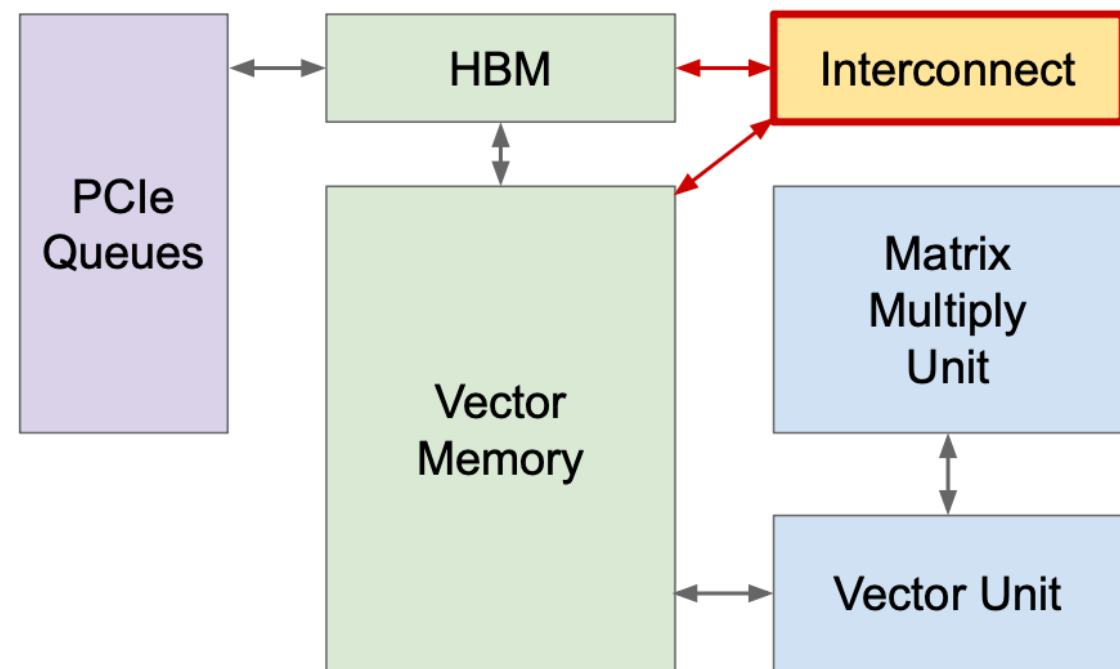
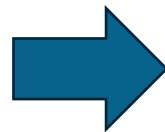
TPUv1架构



TPUv1 -> TPUv2



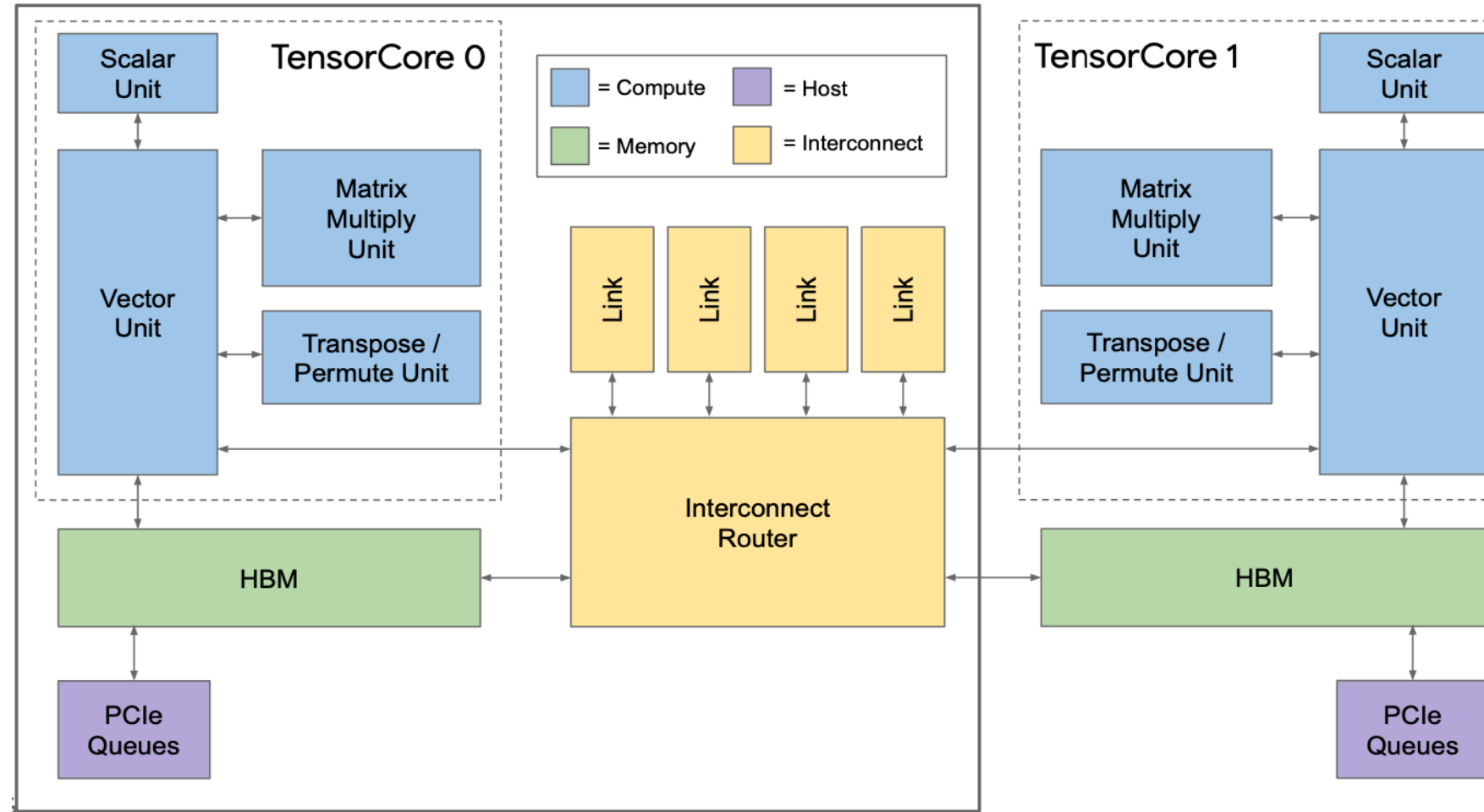
TPUv1: for inference only



TPUv2: for both training and inference

Tensor Core

- VLIW Architecture
- Linear Algebra ISA
 - Scalar, vector, and matrix
 - Built for generality





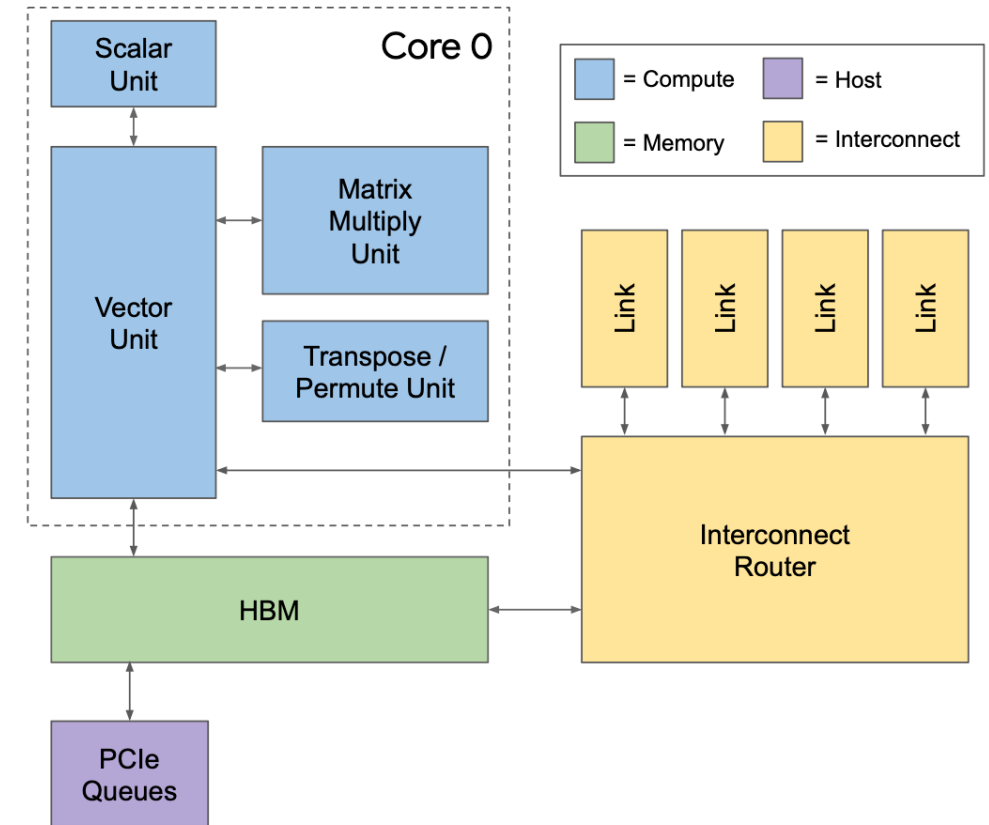
Tensor Core: Scalar Unit

➤ 322b VLIW bundle

- ❑ 2 scalar slots
- ❑ 4 vector slots (2 for load/store)
- ❑ 2 matrix slots (push, pop)
- ❑ 1 misc slot
- ❑ 6 immediates

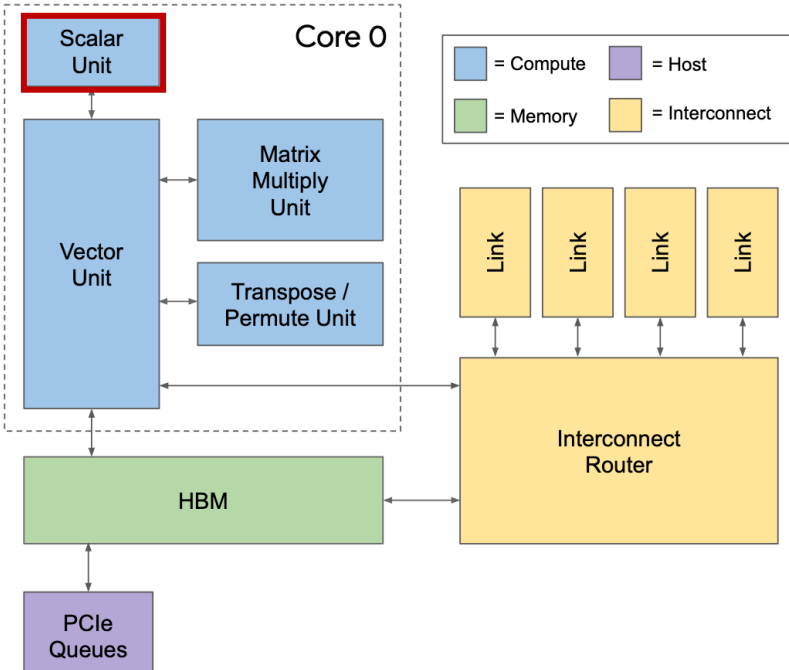
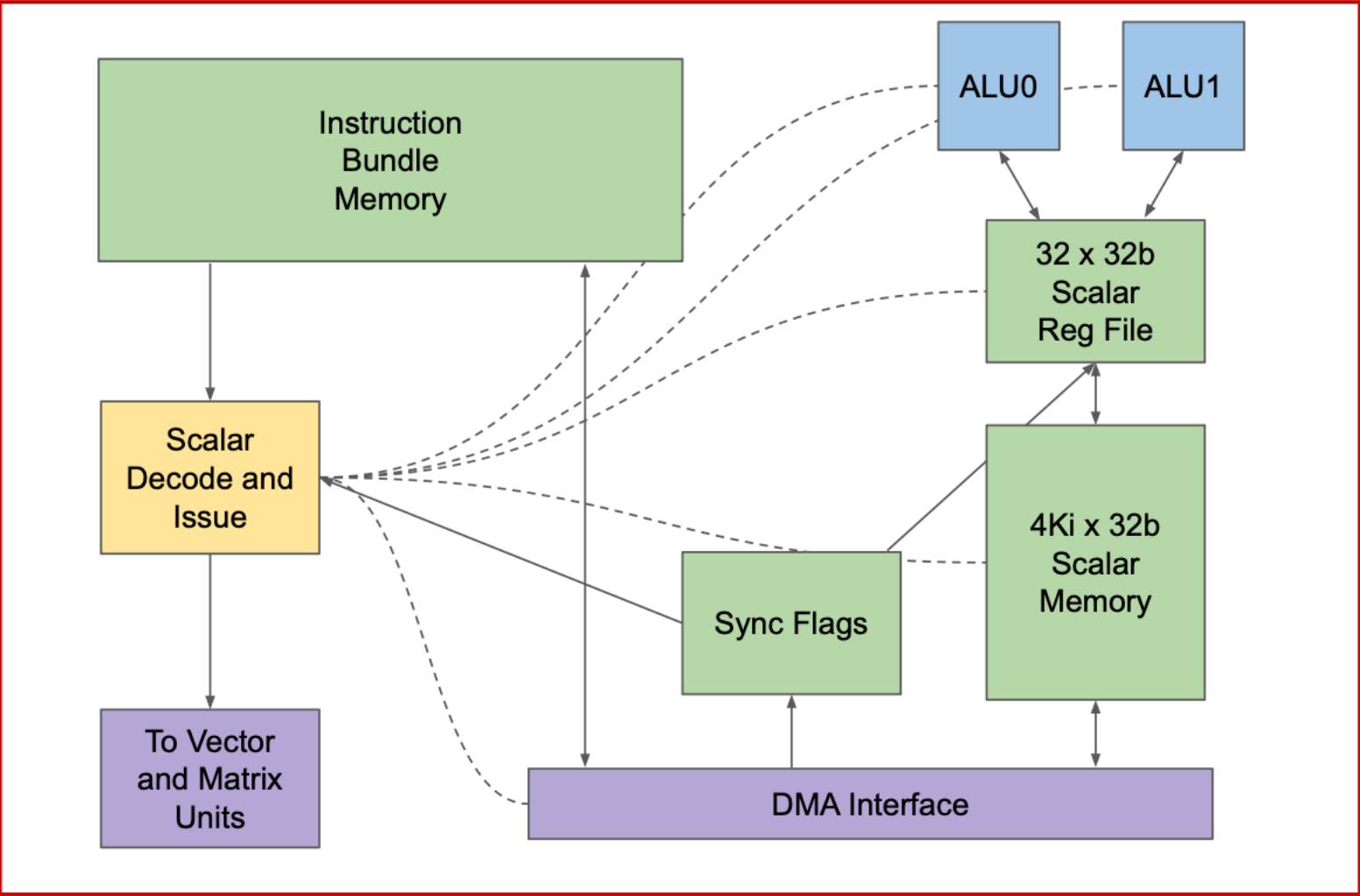
➤ Scalar Unit performs

- ❑ Full VLIW bundle fetch and decode
- ❑ Scalar slot execution



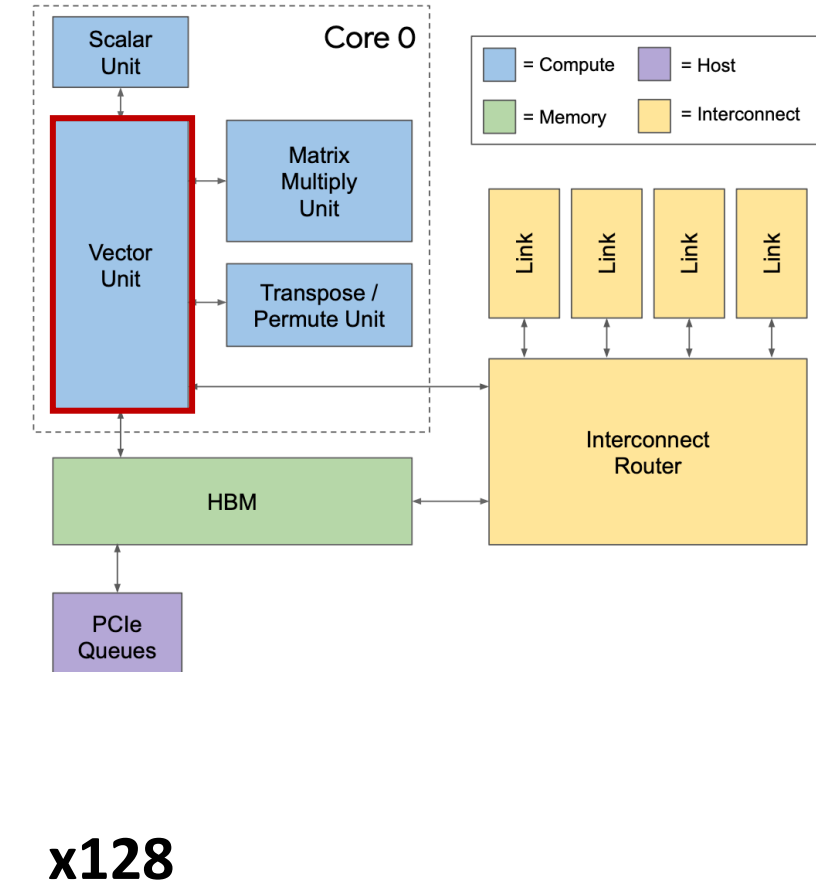
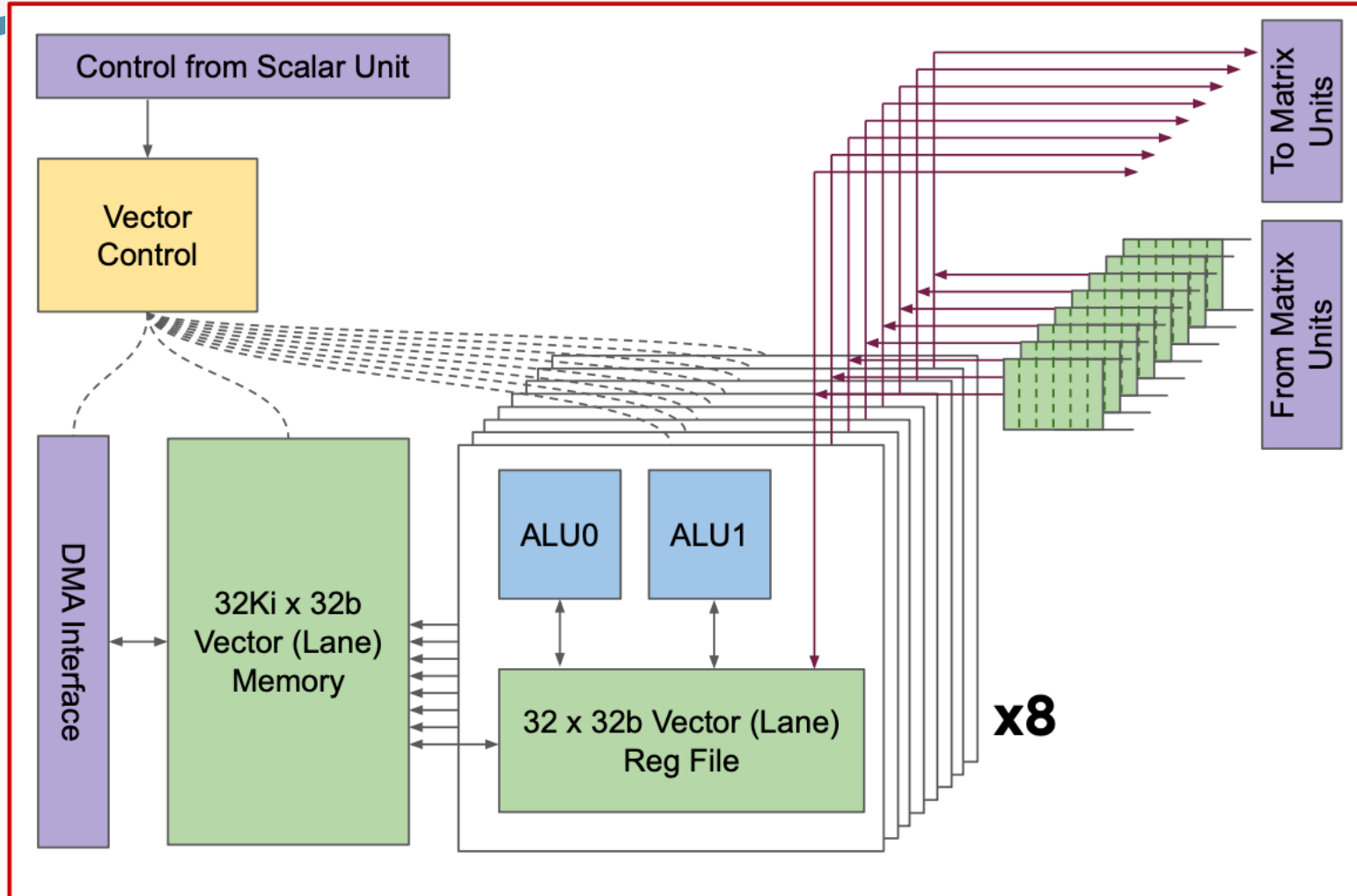


Tensor Core: Scalar Unit





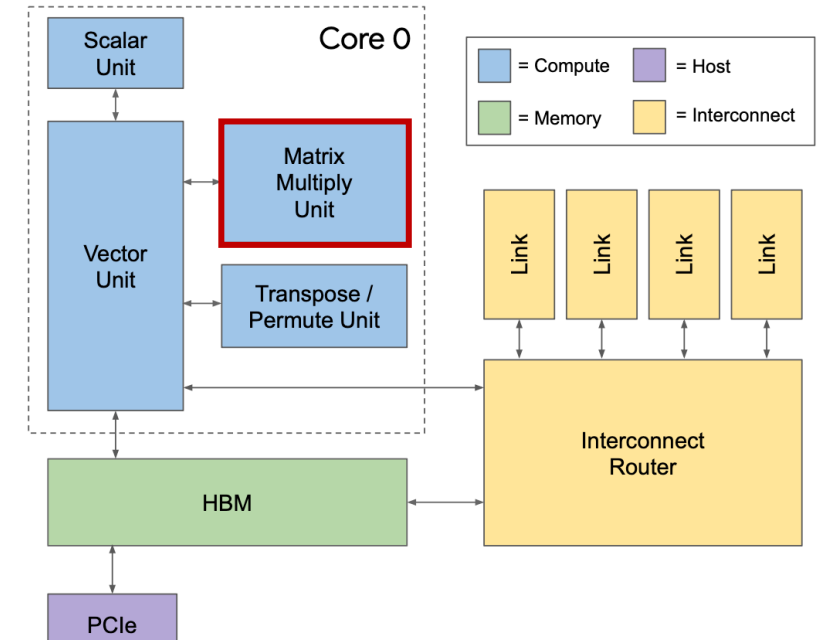
Tensor Core: Vector Unit (Lane)





Tensor Core: Matrix Multiply Unit (MXU)

- 128 x 128 systolic array
 - ❑ Streaming LHS and results
 - ❑ Stationary RHS (w/ optional transpose)
- Numerics
 - ❑ bfloat16 multiply
 - $\{s, e, m\} = \{1, 8, 7\}$
 - The original!
 - ❑ float32 accumulation



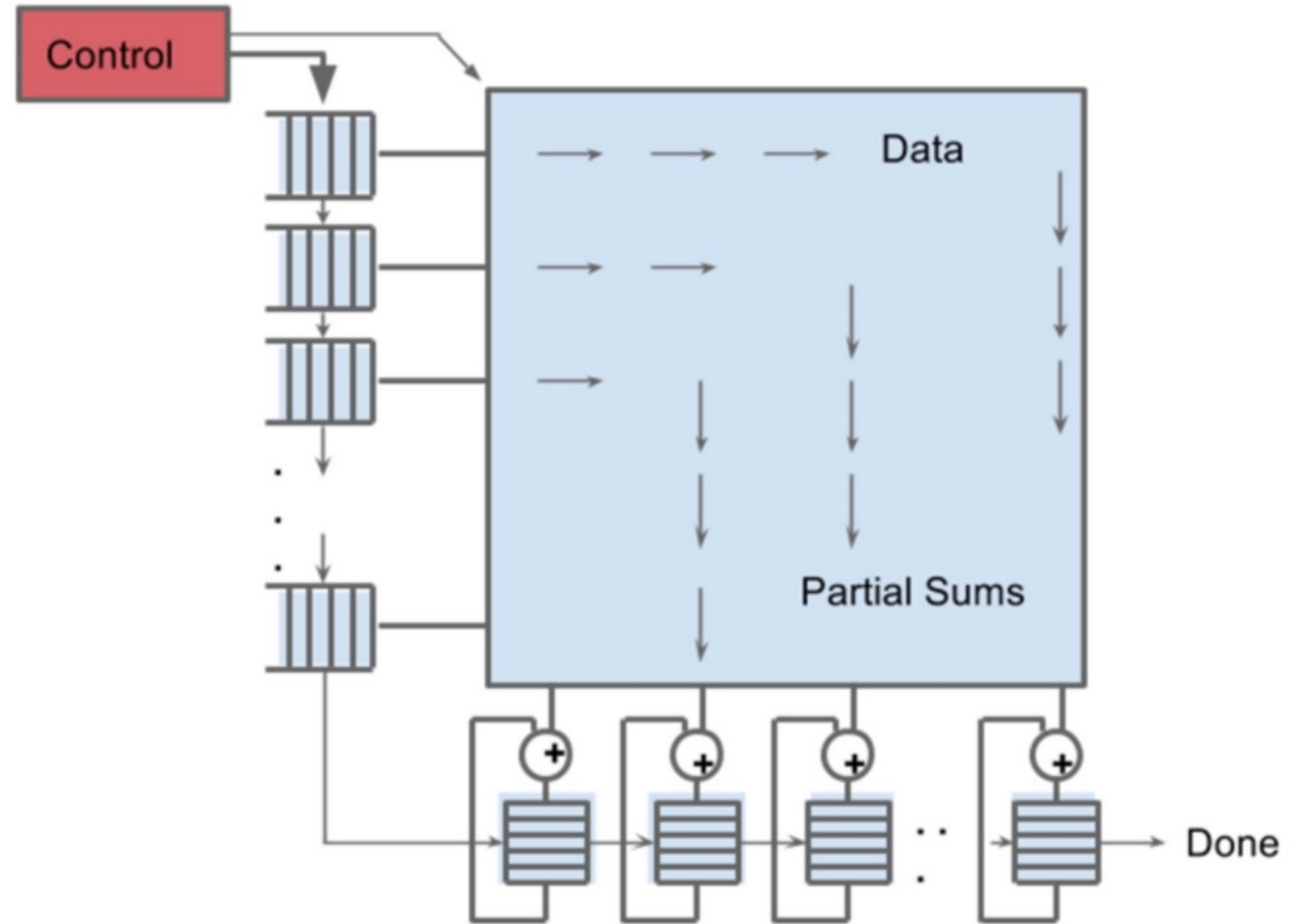
TPU MXU

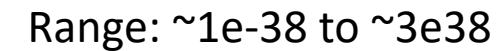


The TPU Matrix Multiplication Unit (MXU) has a **systolic array** mechanism that contains **$256 \times 256 = \text{total } 65,536$ ALUs**.

That means a TPU can process **65,536 multiply-and-adds for 8-bit integers every cycle**.

Because a TPUv1 runs at 700MHz, a TPU can compute **$65,536 \times 700,000,000 = 46 \times 10^{12}$ multiply-and-add operations or 92 Teraops per second (92×10^{12})** in the matrix unit.

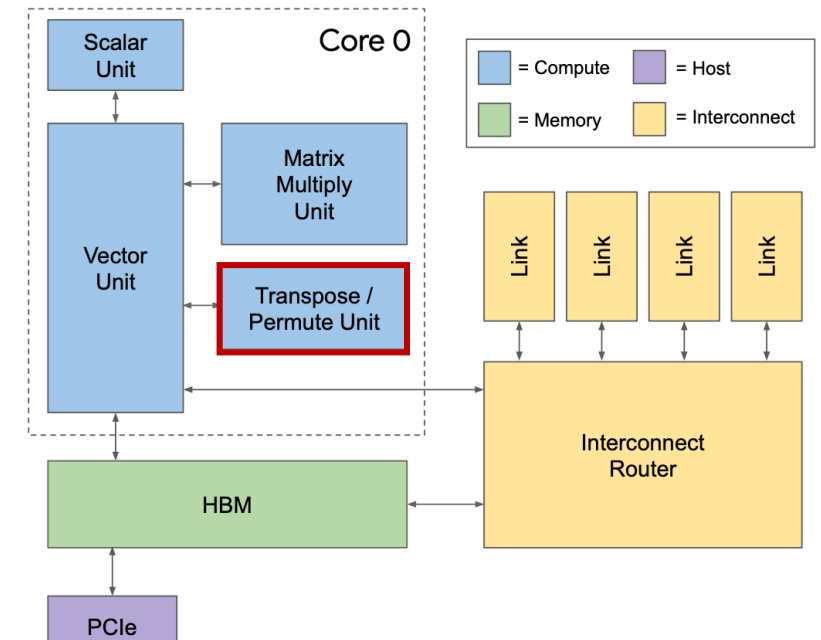






Tensor Core: Transpose, Reduction, Permute Unit

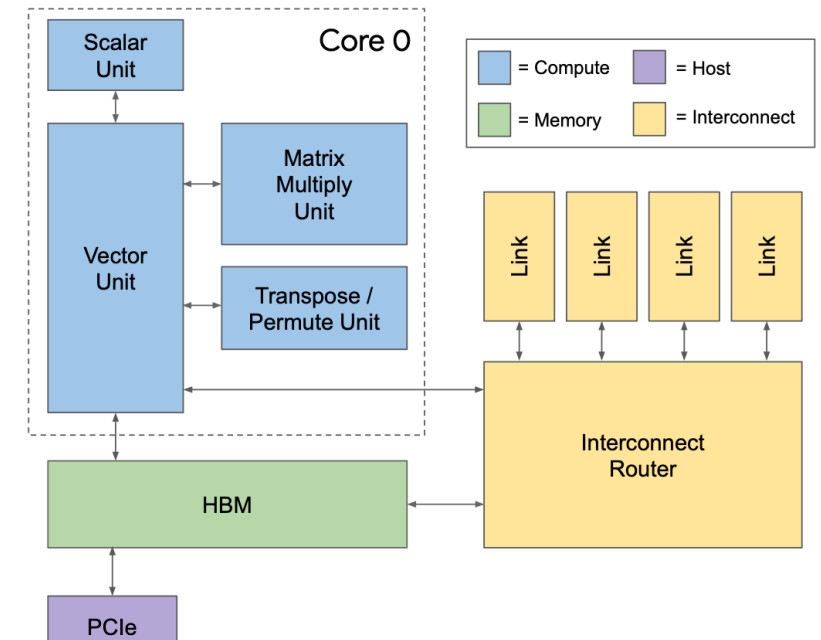
- Efficient common matrix transforms
 - ❑ Transpose
 - ❑ Reduction
 - ❑ Permutation
- Generally, allow reshuffling of data across vector lanes



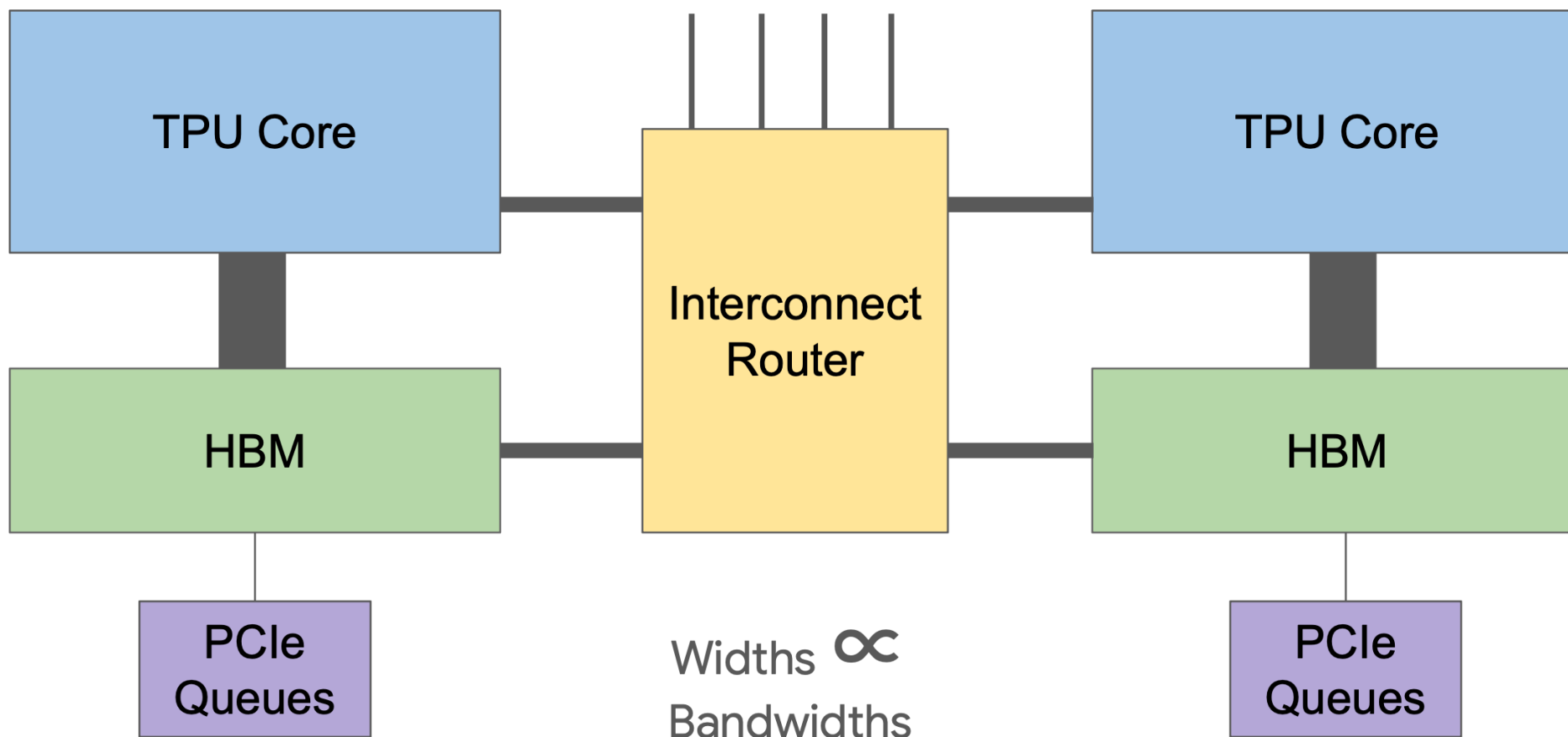
TPU: Memory System



- Loads and stores against SRAM scratchpads
- Provides predictable scheduling within the core
- Can stall on sync flags
- Accessible through asynchronous DMAs
- Indicate completion in sync flags



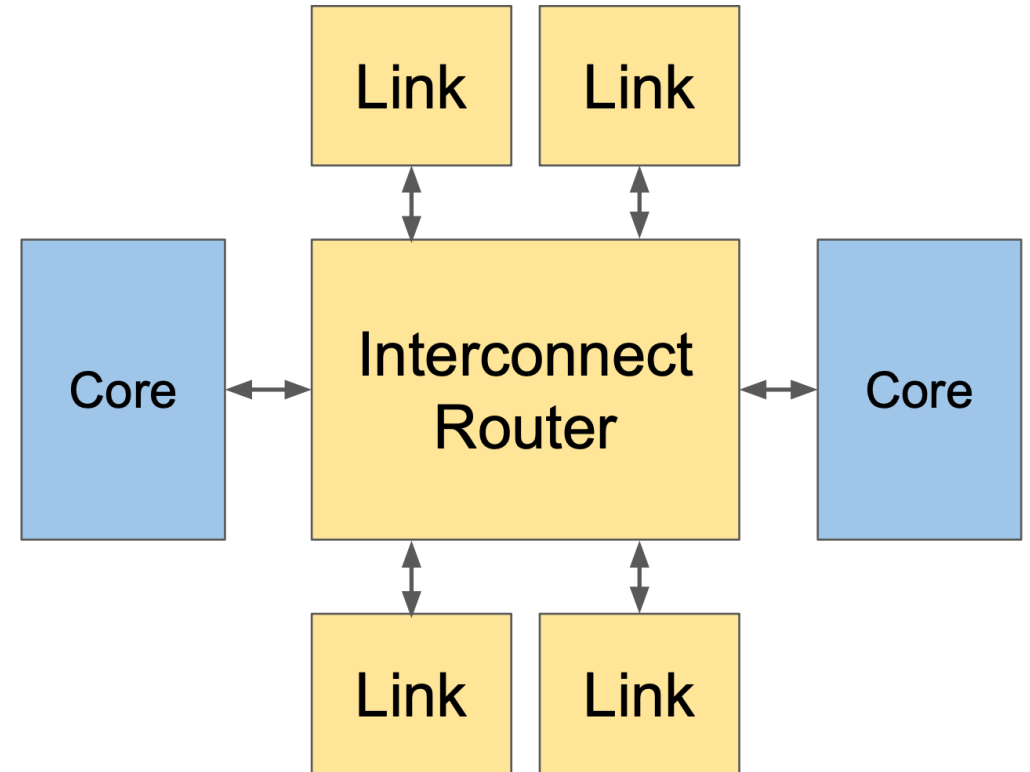
TPU: Memory System across TPUs



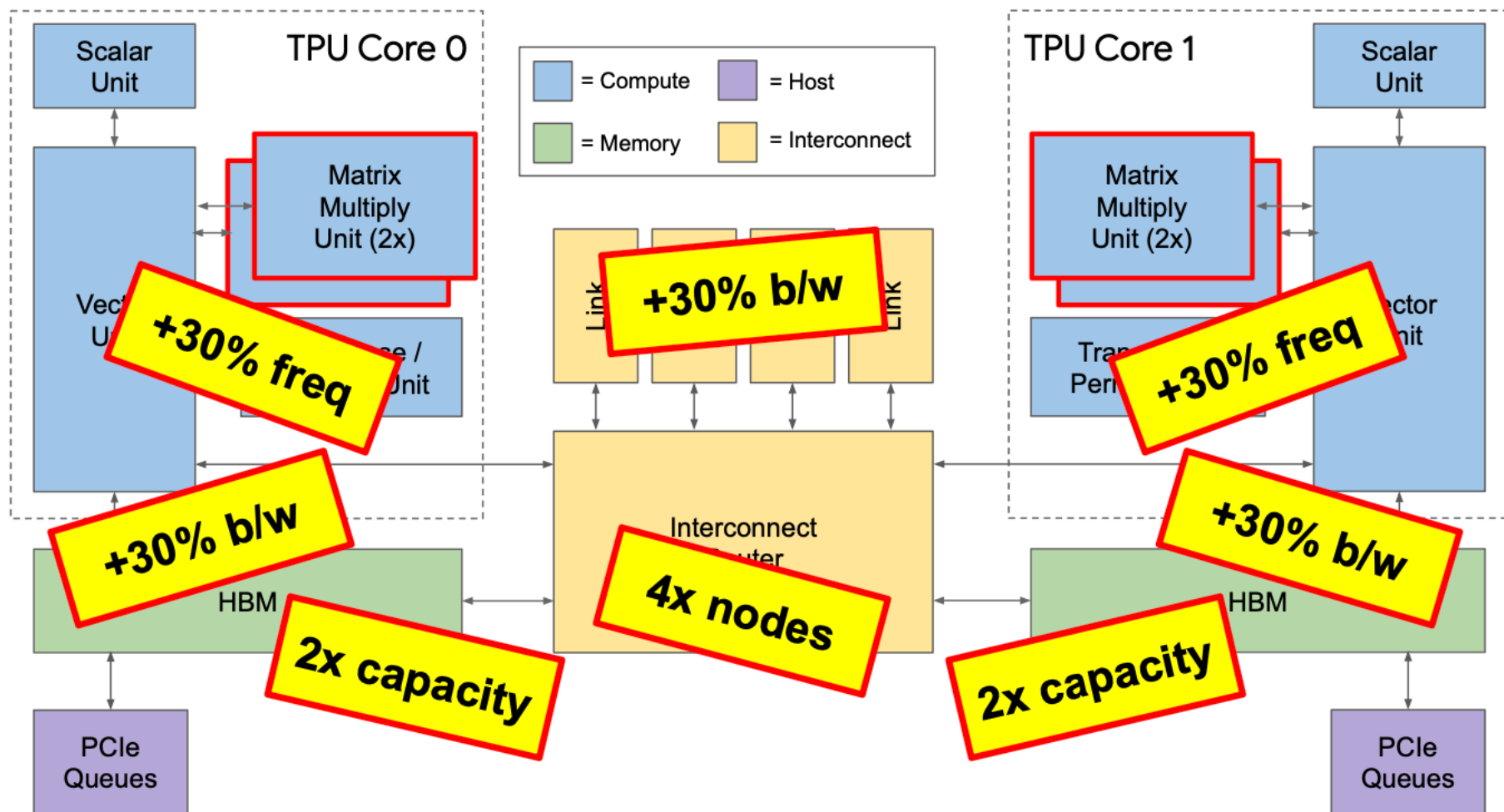
TPU: Interconnect



- On-die router with 4 links
- 500 Gbps per link
- Assembled into 2D torus
- Software view:
 - ❑ Uses DMAs just like HBM
 - ❑ Restricted to push DMAs
 - ❑ Simply target another chip id



TPUv2 -> TPUv3...



Google TPU Programming

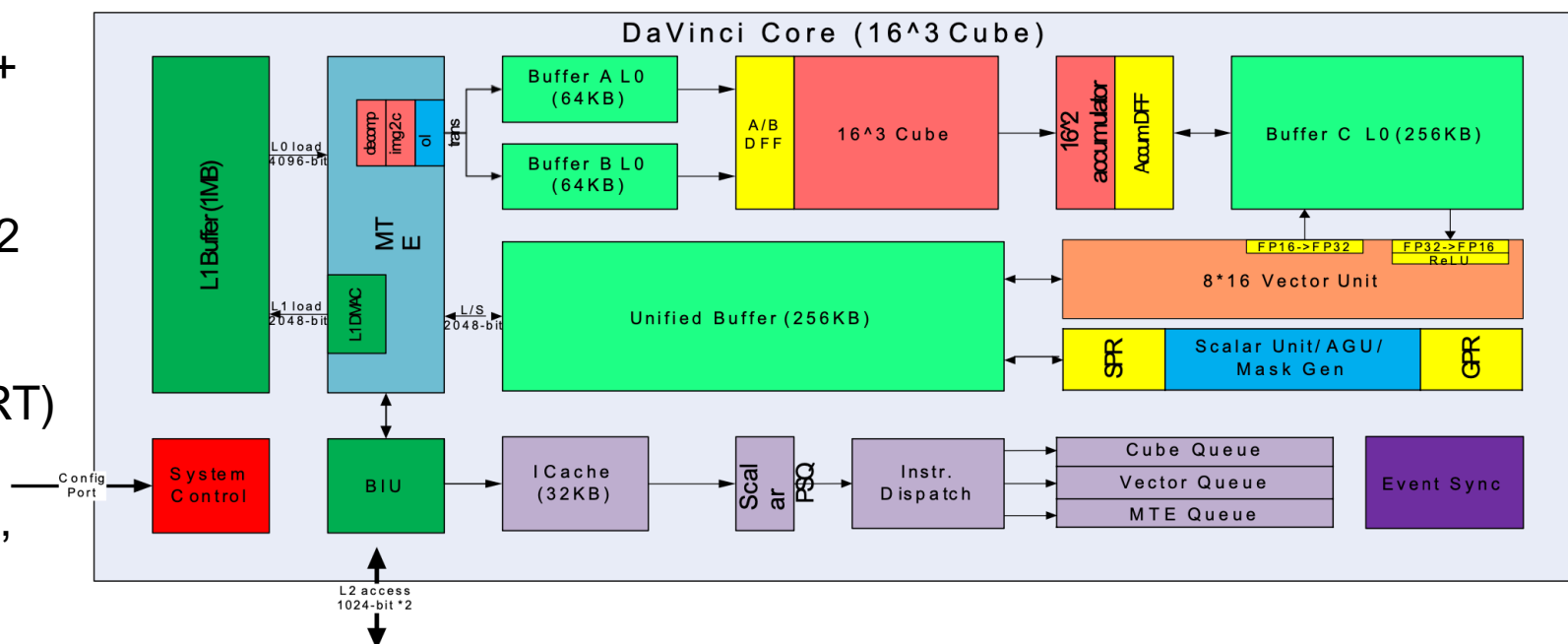


- Google TensorFlow
- Google JAX
- Meta PyTorch
 - w/ PyTorch/XLA

2、Huawei Ascend Neural Processing Unit (NPU)

- Ascend NPU是华为研发的专用于神经网络的处理芯片
- Ascend 910A采用达芬奇架构 (DaVinci) , 于2019年发布

- **Cube:** 4096(16^3) FP16 MACs + 8192 INT8 MACs
- **Vector:** 2048bit INT8/FP16/FP32 vector with special functions (activation functions, NMS-Non Minimum Suppression, ROI, SORT)
- Explicit memory hierarchy design, managed by MTE



Huawei Ascend NPU first Generation



Ascend 310

High Power Efficiency

- Ascend-Mini
- Architecture: DaVinci
- FP16: 8 TeraFLOPS
- INT8 : 16 TeraOPS
- 16 Channel Video Decode – H.264/265
- 1 Channel Video Encode – H.264/265
- Power: 8W
- Process: 12nm



Ascend 910

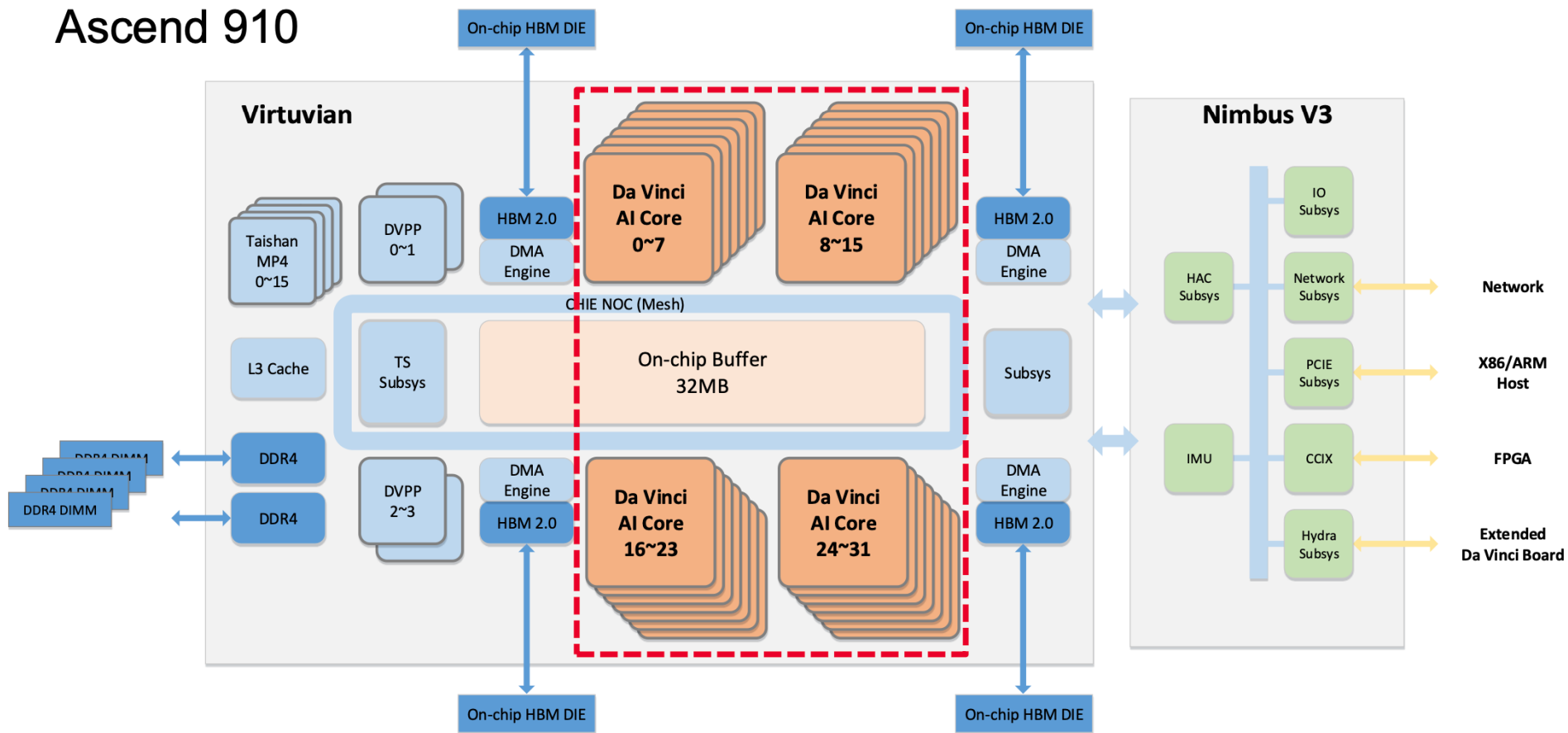
High Computing Density

- Ascend-Max
- Architecture: DaVinci
- FP16: 256 TeraFLOPS
- INT8: 512 TeraOPS
- 128 Channel Video Decode – H.264/265
- Power: 300W
- Process: 7+ nm EUV

Huawei Ascend NPU 910A



Ascend 910



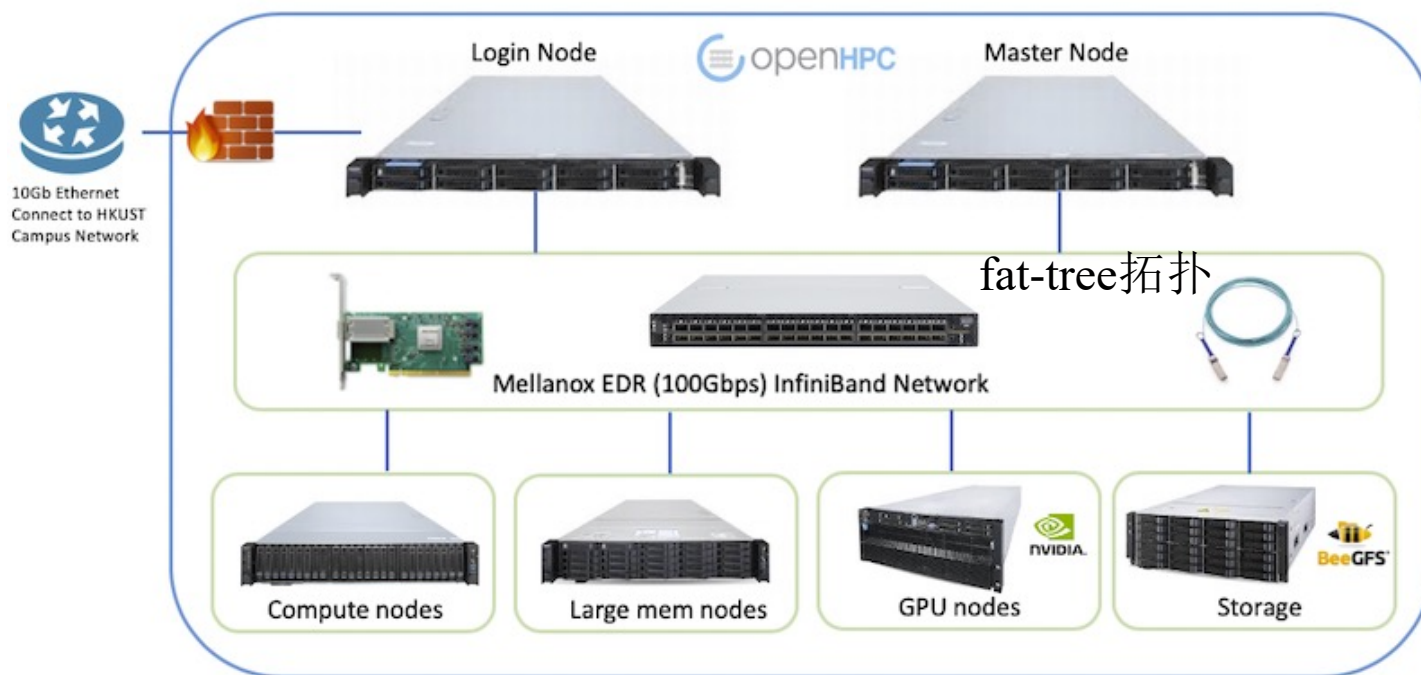
Huawei Ascend NPU Programming



- Huawei CANN (something like CUDA)
- Ascend C programming
 - ❑ Like CUDA C/C++
- MindSpore
 - ❑ Deep learning framework
- Pytorch
 - ❑ <https://github.com/Ascend/pytorch>

3、大规模集群

- ▶ 单个节点
 - ▶ 处理器: e.g., CPU, GPU, TPU, NPU, etc.
 - ▶ 内存: e.g., DRAM, HBM
 - ▶ 外存: HDD, SSD, etc.
 - ▶ 互联: QPI, UPI, PCIe, NVLink, ...
- ▶ 集群
 - ▶ 节点互联:
 - ▶ 以太网 或 InfiniBand
 - ▶ 交换网络
 - ▶ 互联拓扑
 - ▶ Fat-tree, BCube, etc.
 - ▶ 管理节点
 - ▶ 登录, 监看, ...



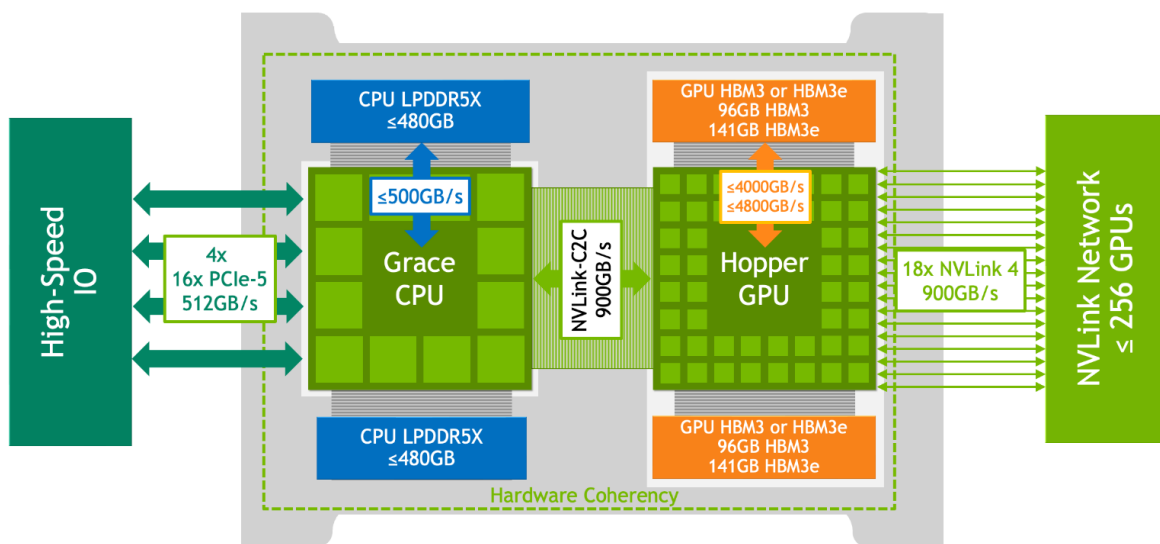
扩展阅读: NVSwitch和NVLink-Network Switch

<https://zhuanlan.zhihu.com/p/678340626>

NVIDIA DGX GH200

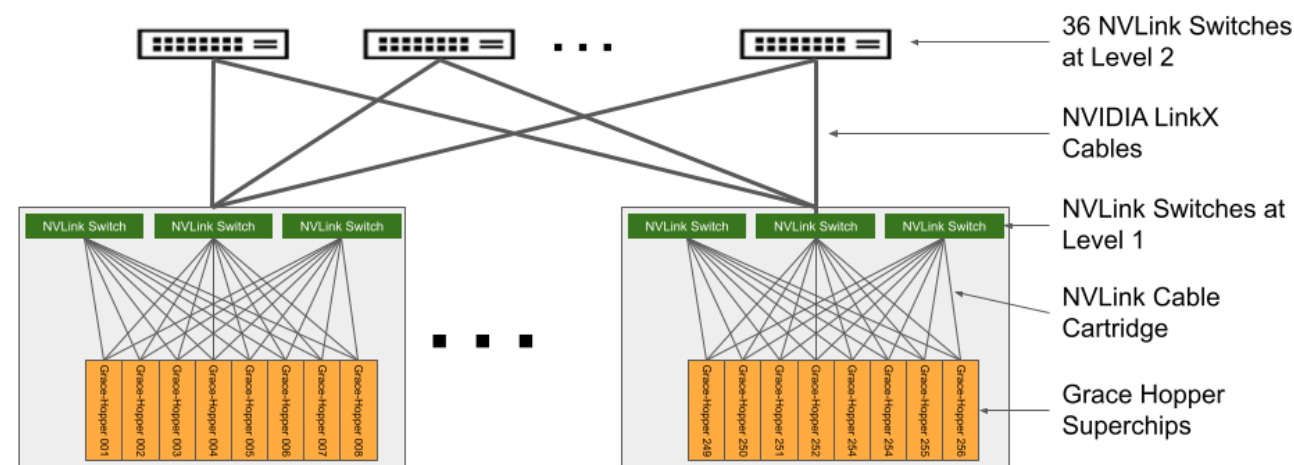


- GH200 Grace Hopper Superchip
- DGX GH200 Supercomputer



Feature	Description
Grace CPU cores (number)	Up to 72 cores
CPU LPDDR5X bandwidth (GB/s)	Up to 500GB/s
GPU HBM bandwidth (GB/s)	4TB/s HBM3 4.9TB/s HBM3e
NVLink-C2C bandwidth (GB/s)	900GB/s total, 450GB/s per direction
CPU LPDDR5X capacity (GB)	Up to 480GB
GPU HBM capacity (GB)	96GB HBM3 144GB HBM3e
PCIe Gen 5 Lanes	64x

Fully Connected NVLink across 256 GPUs

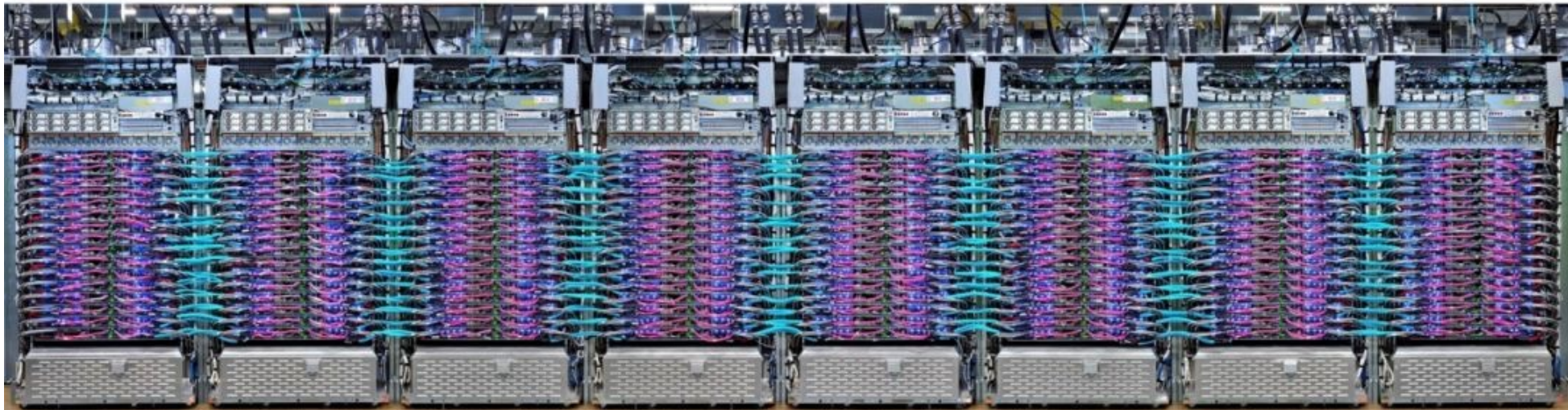


Google TPU Cluster

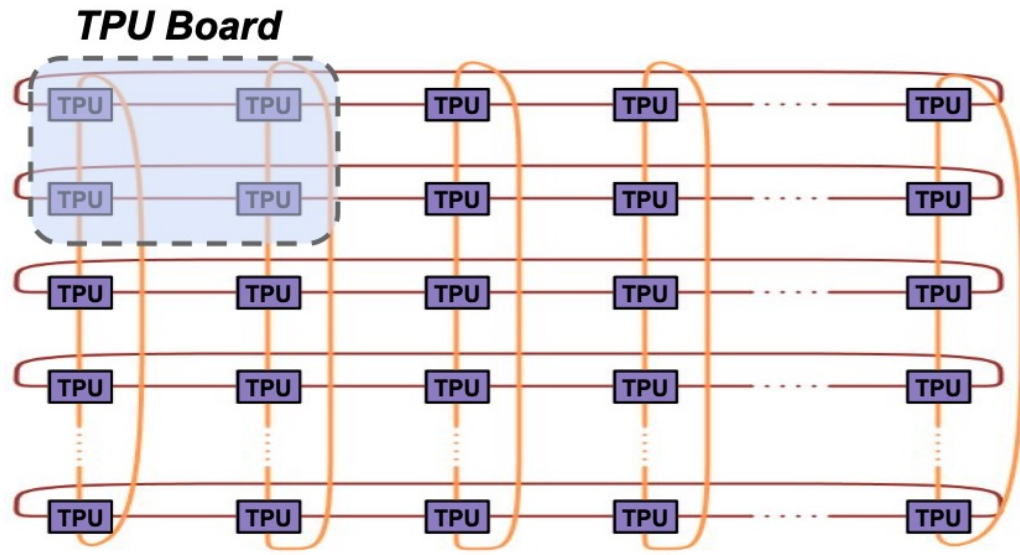


Google TPUv2 Pod (2017)
11.5 petaflops
4 TB HBM
2-D toroidal mesh network
Training and inference

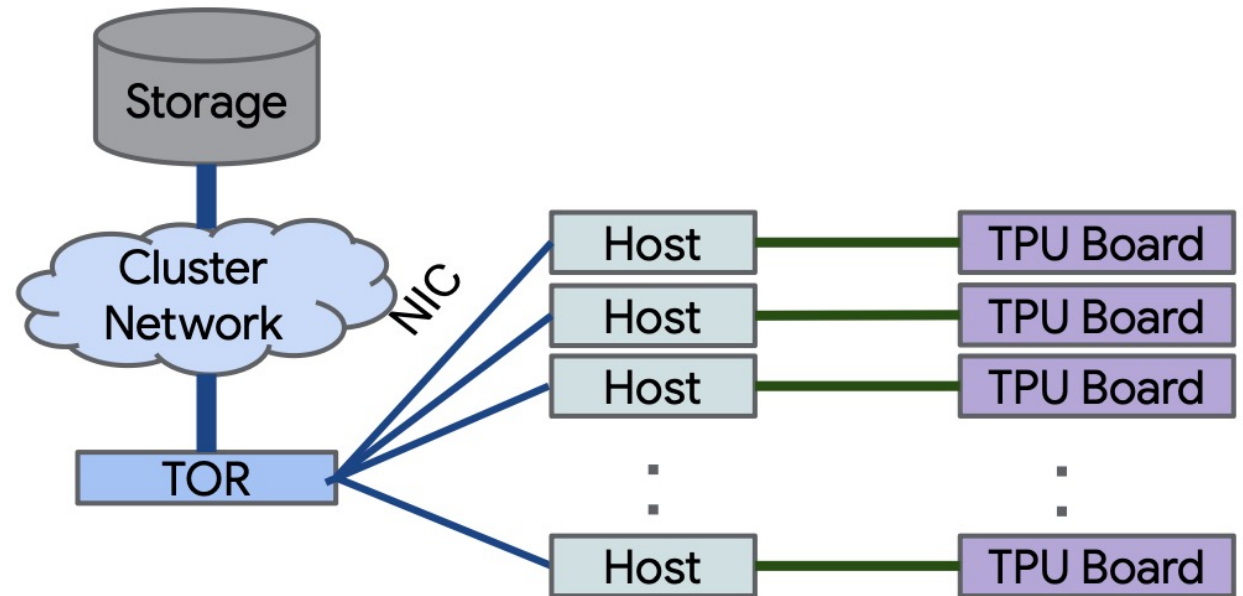
Google TPUv3 Pod (2018)
>100 petaflops



Google TPU Cluster



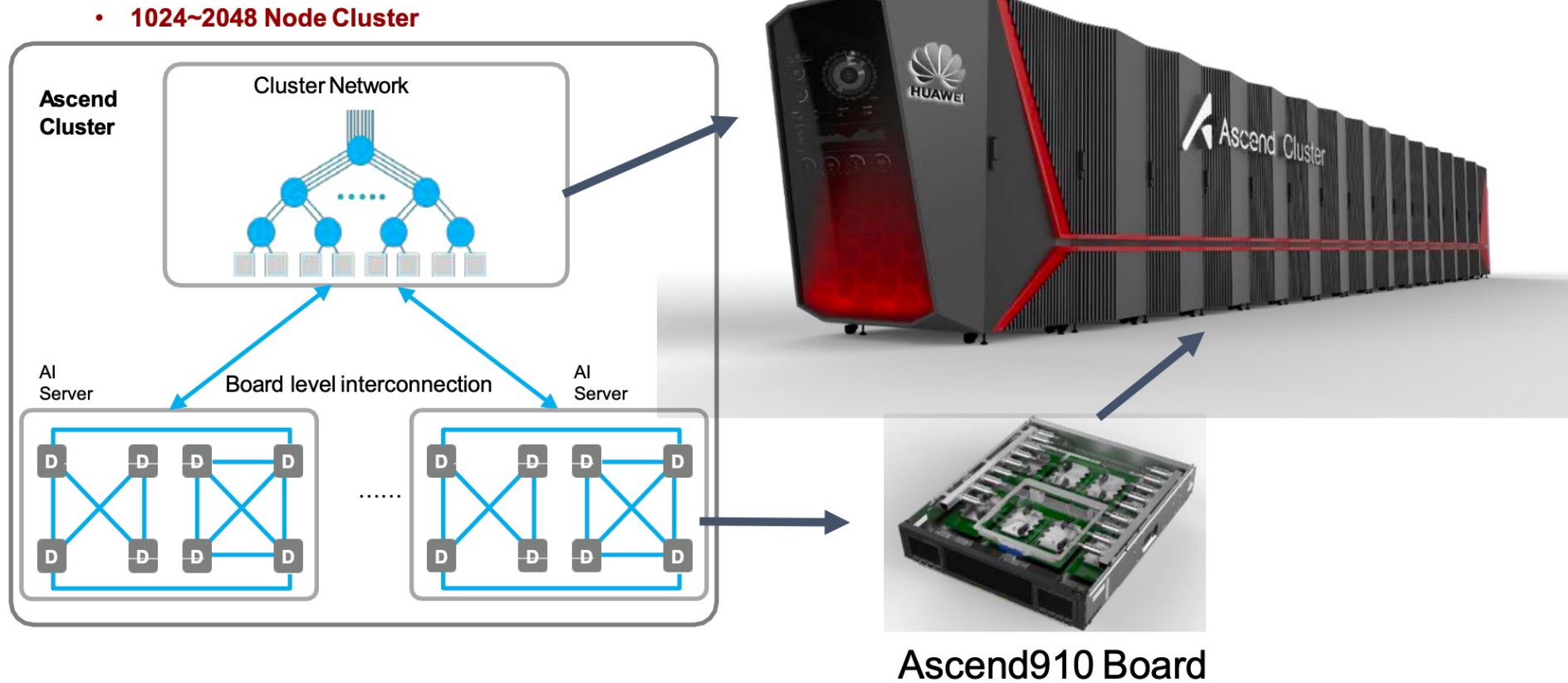
TPUs interconnected in 2D Torus



Ascend 910 Cluster



- 2048 Nodes x 256TFLOPS = 512 Peta FLOPS
- 鹏城云脑II：4096 Nodes



Acknowledgements



Sources:

- https://www.hc32.hotchips.org/assets/program/conference/day2/HotChips2020_ML_Training_Google_Norrie_Patil.v01.pdf
- <https://www.cmc.ca/wp-content/uploads/2020/03/Zhan-Xu-Huawei.pdf>



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谢谢

Thank You